

Tregopathies: Monogenic diseases resulting in regulatory T-cell deficiency



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1. To understand the development, signaling, and functions of regulatory T (Treg) cells.
2. To define the immunopathogenic mechanisms of Treg cell deficiencies.
3. To diagnose monogenic diseases that affect Treg cell functions.
4. To learn about treatment options for Treg cell deficiencies.

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Monogenic diseases of the immune system, also known as inborn errors of immunity, are caused by single-gene mutations resulting in immune deficiency and dysregulation. More than 350 diseases have been described to date, and the number is rapidly expanding, with increasing availability of next-generation sequencing facilitating the diagnosis. The spectrum of immune dysregulation is wide, encompassing deficiencies in humoral, cellular, innate, and adaptive immunity; phagocytosis; and the complement system, which lead to autoinflammation and autoimmunity. Multiorgan autoimmunity is a dominant symptom when genetic mutations lead to defects in molecules

essential for the development, survival, and/or function of regulatory T (Treg) cells. Studies of “Tregopathies” are providing critical mechanistic information on Treg cell biology, the role of Treg cell-associated molecules, and regulation of peripheral tolerance in human subjects. The pathogenic immune networks underlying these diseases need to be dissected to apply and develop immunomodulatory treatments and design curative treatments using cell and gene therapy. Here we review the pathogenetic mechanisms, clinical presentation, diagnosis, and current and future treatments of major known Tregopathies caused by mutations in *FOXP3*, *CD25*, cytotoxic

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T lymphocyte–associated antigen 4 (CTLA4), LPS-responsive and beige-like anchor protein (LRBA), and BTB domain and CNC homolog 2 (BACH2) and gain-of-function mutations in signal transducer and activator of transcription 3 (STAT3). We also discuss deficiencies in genes encoding STAT5b and IL-10 or IL-10 receptor as potential Tregopathies. (J Allergy Clin Immunol 2018;142:1679-95.)

Key words: Autoimmunity, BACH2, CD25, CTLA-4, FOXP3, gain of function, hematopoietic stem cell transplantation, IL-10, IL-2, immune dysregulation, inborn error of immunity, IPEX, LRBA, monogenic disease, primary immunodeficiency, regulatory T (Treg) cell, STAT3 GOF, T_R1 , Tregopathy, type 1 regulatory T cell

Regulatory T (Tregs) cells are central players in the maintenance of immune homeostasis and immune tolerance. In 2017, the International Union of Immunological Societies (IUIS) updated the classification of inborn errors of immunity.¹ Treg cell defects fall within the category of diseases of immune dysregulation and currently encompass diseases resulting from loss-of-function (LOF) mutations in *FOXP3*, *CD25*, cytotoxic T lymphocyte–associated antigen 4 (*CTLA4*), LPS-responsive and beige-like anchor protein (*LRBA*), and BTB domain and CNC homolog 2 (*BACH2*) and a gain-of-function (GOF) mutation in signal transducer and activator of transcription 3 (*STAT3*). Each of these genes encode proteins that uniquely contribute to the function of thymus-derived forkhead box P3 (*FOXP3*)⁺ regulatory T (tTreg) cells (Fig 1, A). Certain proteins, including CTLA-4, exhibit regulatory functions in different Treg cell subsets, such as peripheral inducible IL-10–producing type 1 regulatory T (T_R1) cells (Fig 1, B) or in activated effector T (Teff) cells. Mutations in genes impairing T-cell development, such as *RAG1*, *RAG2*, or *CD3G*, also lead to Treg cell deficiency and are associated with immune dysregulation.^{2,3} Although less common, some immunodeficiencies present with low frequencies of Treg cells, which are still functional, as in patients with 22q11 deletion (our unpublished data).^{4,5}

These different types of Treg cell–dependent immune dysregulations often display a similar clinical presentation but might require different treatment. Therefore an in-depth characterization of the basic biology of the defects that cause “Tregopathies” is desirable to advance their diagnosis and treatment.

In this review, we focus on the pathogenetic mechanisms underlying the 6 IUIS-classified Tregopathies. We speculate on whether the immune dysregulation might also stem from alteration of T_R1 -related molecules. In addition, we highlight the importance of early diagnosis and which treatments targeting Treg cell–dependent immune dysregulation are available or under development.

MOLECULAR IDENTITY AND FUNCTION OF TREG CELLS

FOXP3⁺ Treg cells

Treg cells are specialized T cells responsible for the maintenance of peripheral self-tolerance and regulation of immune responses both in mice and human subjects. Treg cells suppress various Teff cell functions, including proliferation, production of proinflammatory cytokines and growth factors, and expression of costimulatory molecules. Treg cells also

Abbreviations used

allo-HSCT:	Allogeneic hematopoietic stem cell transplantation
APC:	Antigen-presenting cell
BACH2:	BTB domain and CNC homolog 2
CTLA-4:	Cytotoxic T lymphocyte–associated antigen 4
GH:	Growth hormone
GOF:	Gain of function
HDR:	Homology-directed repair
HSC:	Hematopoietic stem cell
IBD:	Inflammatory bowel disease
IL-10R:	IL-10 receptor
IPEX:	Immunodysregulation, polyendocrinopathy, enteropathy, X-linked
IUIS:	International Union of Immunological Societies
LOF:	Loss of function
LRBA:	LPS-responsive and beige-like anchor protein
NK:	Natural killer
PD-1:	Programmed cell death protein 1
pTreg:	Peripherally induced regulatory T
SCID:	Severe combined immunodeficiency
SLE:	Systemic lupus erythematosus
SOCS3:	Secretion of cytokine signaling 3
STAT:	Signal transducer and activator of transcription
T1D:	Type 1 diabetes
Teff:	Effector T
T_R1 :	Inducible IL-10–producing type 1 regulatory T
Treg:	Regulatory T
TSDR:	Regulatory T cell–specific demethylated region
tTreg:	Thymus-derived FOXP3 ⁺ regulatory T

regulate the activation of other cell subsets, including monocytes and B cells.⁶⁻⁹

Several subsets of Treg cells exist, each characterized by unique features, as well as shared molecules and mechanisms of action. tTreg cells, which originate from the thymus,^{10,11} were originally described to maintain self-tolerance in mice¹² and are characterized by the CD4⁺CD25 (IL-2 receptor α chain)^{high}CD127 (IL-7 receptor)[–]FOXP3⁺ phenotype.¹³ tTreg cells are present at low frequencies, accounting for approximately 5% of total CD4⁺ T cells in peripheral blood.¹⁴ Although the absolute number of tTreg cells decreases from infancy to adulthood (consistent with decreased lymphopoiesis throughout development), their frequency relative to CD4⁺ T cells remains steady.¹⁴

Several models have proposed self-reactive T-cell receptor affinity to explain the generation of Treg cells in the thymus, although the precise mechanism of origin is still debatable. The transcription factor FOXP3^{15,16} is essential for the regulatory function of tTreg cells. Mutations in *FOXP3* cause severe autoimmunity both in mice and human subjects, and the relevance of FOXP3 to immune regulation was further elucidated by gene knockout specifically targeting Treg cells.¹⁷⁻¹⁹ Also, tTreg cells have a unique and exclusive pattern of DNA demethylation in the promoter/enhancer regions of *FOXP3* (regulatory T cell–specific demethylated region [TSDR], also known as CNS2). Demethylation of the *FOXP3* promoter and TSDR, along with acquisition of activating histone modifications in conserved elements, confers stable *FOXP3* expression and regulatory function. Therefore the most precise method to count tTreg cells is through quantification of TSDR demethylation.^{20,21}

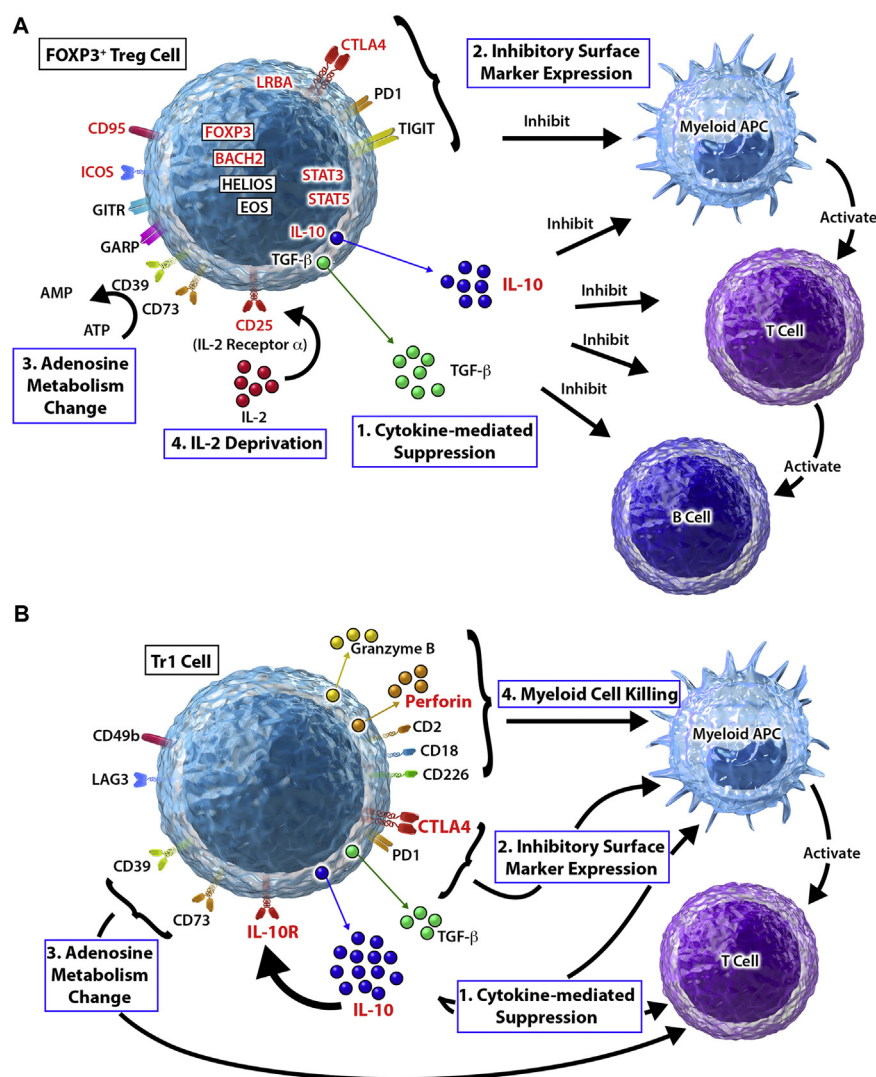


FIG 1. A, Mechanisms and molecules necessary for the suppressive function of FOXP3⁺ Treg cells. FOXP3⁺ Treg cells are CD4⁺ T cells phenotypically defined by their CD25^{high}CD127^{low}FOXP3^{high} expression. FOXP3⁺ Treg cells also express inhibitory molecules on the surface (CTLA-4, PD-1, and TIGIT) and secrete immunosuppressive cytokines (IL-10 and TGF- β) through which they exert their regulatory function. A number of other surface markers (CD39, CD73, GARP, GITR, ICOS, and CD95) and transcription factors (BACH2, HELIOS, and EOS) also play a functional role in FOXP3⁺ Treg cells. FOXP3⁺ Treg cells suppress their target cells through various mechanisms, such as (1) secretion of immunosuppressive cytokines that suppress both APC and T cell activation and inhibit T-cell proliferation (inhibitory cytokines can exert the same effect on B cells); (2) expression of inhibitory molecules on the surface that inhibit activation of APCs, which present antigen and provide crucial costimulatory signals to T cells; (3) conversion of ATP into AMP to adenosine through CD39/CD73 to alter the local T-cell metabolic environment and stimulate inhibitory receptor expression through adenosine receptors; and (4) IL-2 deprivation from the environment through increased consumption because FOXP3⁺ Treg cells express constitutively high levels of the high-affinity IL-2 receptor CD25. Red molecule names indicate that the mutation of the molecule is reported as a monogenic disease. GARP, Glycoprotein A repetitions predominant; GITR, glucocorticoid-induced TNF receptor family-related protein; ICOS (CD278), inducible T-cell stimulator. **B,** Mechanisms and molecules necessary for suppressive function of Tr1 cells. CD49b⁺LAG3⁺ Tr1 cells are defined by their high expression of IL-10 in the absence of other T cell-polarizing cytokines (IL-2, IL-4, and IL-17). IL-10 is crucial to Tr1 cells as both an effector mechanism of immune suppression and a key signaling molecule to induce the phenotype. Tr1 cells possess a variety of immunosuppressive mechanisms, such as (1) secretion of the immunosuppressive cytokines IL-10 and TGF- β , which can act directly on T cells by inhibiting proliferation and secretion of proinflammatory cytokines (these cytokines also act on professional APCs by causing downregulation of MHC class II and costimulatory molecules to prevent T-cell activation, as well as polarizing APCs toward adopting tolerogenic phenotypes and secreting immunosuppressive cytokines); (2) expression of inhibitory surface molecules, such as CTLA-4 and PD-1, which have been shown to inhibit T-cell activation through cell contact-dependent mechanisms; (3) conversion of ATP into AMP to adenosine through CD39/CD73 to alter the local T-cell metabolic environment and stimulate inhibitory receptor expression through adenosine receptors; and (4) killing of myeloid cells, including several APC lineages, through a granzyme B/perforin-mediated mechanism shown to require CD2/CD18/CD226 surface expression to prevent T-cell activation. Red molecule names indicate that the mutation of the molecule is reported as a monogenic disease.

FOXP3 maintenance in Treg cells and its induction in other T-cell subsets depend on TCR-mediated activation of the mammalian target of rapamycin (mTOR)/AKT/phosphoinositide 3-kinase signaling pathway and phosphorylation of STAT5²² induced by IL-2, IL-7, and IL-15. tTreg cells are also phosphorylated STAT5⁺, and are able to respond to low levels of IL-2.²²⁻²⁴ FOXP3⁺ Treg cells exert their immunosuppressive functions in various ways. FOXP3 suppresses genes related to cell cycle and cytokines (eg, *IL2*, *IL4*, and *IFNG*). Notably, these effects occur constitutively in tTreg cells but also in Teff cell that express low and transient levels of FOXP3 on activation. Similarly, constitutive FOXP3 expression in tTreg cells also mediates upregulation of CD25 and CTLA-4, which are essential for their survival and function, respectively. Treg cells express a number of T-cell coinhibitory surface molecules, such as CTLA-4, programmed cell death protein 1 (PD-1), and T cell immunoreceptor with Ig and ITIM domains (TIGIT), which extrinsically inhibit T-cell proliferation and effector functions by repressing costimulatory and TCR signaling.²⁵ Treg cells can also inhibit immune responses by (1) secreting the suppressive cytokines IL-10 and TGF- β ,²⁶ (2) hydrolyzing ATP through CD39 ectonucleotidase,^{27,28} or (3) inhibiting Teff cell proliferation by consuming IL-2 through constitutive expression of CD25, the high-affinity IL-2 receptor.²⁹ In addition, Treg cells can directly suppress B cells,³⁰ but the precise mechanism is not fully understood. Therefore mutations in genes critical for Treg cell function can variably affect the Treg cell-dependent regulatory network and manifest clinically as immune dysregulation.

It is now well accepted that FOXP3⁺ Treg cells can also differentiate extrathymically in peripheral tissues (peripherally induced regulatory T [pTreg] cells) or can be induced *in vitro* in the presence of TGF- β .³¹ Unlike tTreg cells, pTreg cells do not possess a distinctive epigenetic signature,³² making their identification in human peripheral blood challenging.³³ T_R1 cells represent a FOXP3-independent subset of peripheral inducible Treg cells³⁴ that play an important role in regulating antigen-specific tolerance. For the purpose of this review, we will use *Treg cell* for FOXP3⁺ Treg cells and *T_R1 cell* for the inducible, FOXP3-independent type 1 Treg cells, as discussed below.

T_R1 cells

T_R1 cells are Treg cells defined by high expression of IL-10 and capacity to suppress Teff cell responses. T_R1 cells were initially identified in the peripheral blood of patients with severe combined immunodeficiency (SCID) who had long-term tolerance toward allogeneic hematopoietic stem cell transplantation (allo-HSCT).³⁵ Subsequent studies revealed that T_R1 cells play an active role in suppressing local tissue inflammation in healthy subjects. Although a master transcription factor analogous to FOXP3 has not yet been identified in T_R1 cells, *in vivo*- and *in vitro*-derived human and mouse T_R1 cells can be defined by coexpression of the surface markers CD49b and LAG3.³⁶ Our *in vitro* induction protocols show that IL-10, together with prolonged TCR triggering, drives T_R1 differentiation.³⁷⁻³⁹ Additional cytokines and costimulatory molecules, such as IL-27 and CD46, have been shown to polarize CD4⁺ T cells toward a T_R1-like phenotype through upregulation of IL-10 secretion.^{40,41}

TCR activation is required for the suppressive functions of T_R1 cells, resulting in secretion of a characteristic T_R1 cytokine profile (high IL-10 and TGF- β 1 levels, intermediate IFN- γ levels, and low/no IL-2, IL-4, and IL-17 levels).³⁷ T_R1 cells suppress inflammation primarily through secretion of IL-10 and TGF- β 1 in addition to possessing the capacity to kill myeloid antigen-presenting cells (APCs) through perforin and granzyme B-mediated lysis (Fig 1, B).⁴² The mechanisms of suppression between FOXP3⁺ Treg and T_R1 cells were compared previously.⁴³

Although the lack of a master transcription factor has made it difficult to directly observe the effects of T_R1 cell function loss, several studies have investigated T_R1 cells in the context of human autoimmune diseases. CD46-costimulated T_R1 cells derived from patients with multiple sclerosis showed reduced IL-10 secretion compared with those from healthy control subjects.^{44,45} Patients with psoriasis had decreased levels of CD49b⁺LAG3⁺ T_R1 cells, and their T_R1 frequency inversely correlated with disease severity.⁴⁶ In nondiabetic subjects islet autoantigen-specific, IL-10-secreting population of CD4⁺ T cells was found to be suppressive and possess a T_R1-like ability to lyse islet-antigen presenting APCs.⁴⁷ On the other hand, cells from patients with type 1 diabetes (T1D) displayed skewing toward IFN- γ production in response to islet autoantigens.⁴⁸ Furthermore, response to proinsulin peptide immunotherapy in patients with T1D was associated with higher CD4⁺ T-cell IL-10 production,⁴⁹ and higher levels of IL-10-producing cells at diagnosis were linked to better glucose regulation later.⁵⁰ Finally, human intestinal T_R1 cells, identified as the IFN- γ ⁺IL-10⁺CCR5⁺PD-1⁺ subset, displayed reduced IL-10 production in patients with inflammatory bowel disease (IBD), thereby associating an impairment of T_R1 function with IBD in human subjects.⁵¹

MONOGENIC DISEASES THAT AFFECT FOXP3⁺ TREG CELL FUNCTION: CLINICAL AND IMMUNOLOGIC PRESENTATION

Hemizygous mutations in *FOXP3*, which is located on the X-chromosome, manifest in human subjects as immunodysregulation, polyendocrinopathy, enteropathy, X-linked (IPEX) syndrome (a syndrome caused by a deleterious mutation of the *FOXP3* gene).⁵² Typically, patients with IPEX syndrome present with severe enteropathy, T1D, and eczema very early in life, with a 2-month median onset age.⁵³

Recently, we published a comprehensive study including 96 patients with IPEX syndrome that characterized the clinical history of the disease and treatment outcomes.⁵³ More than 50 different mutations have been described in patients along the entire *FOXP3* gene, with enrichment in the FKH domain. The site of mutations does not seem to affect the phenotype, course, or outcome of the disease.⁵³ The disease is difficult to diagnose because it can present as isolated early-onset diabetes⁵⁴ or enteropathy⁵⁵ or manifest later in childhood.⁵⁶ In addition to the predominant triad of symptoms and failure to thrive caused by enteropathy and chronic illness, other autoimmune manifestations have been observed, including nephropathy, hepatitis, autoimmune thyroiditis, autoimmune hemolytic anemia, thrombocytopenia, neutropenia, lymphadenopathy, arthritis, alopecia, food allergies, and, rarely, neurologic manifestations.⁵³ Gut biopsy specimens at diagnosis are

characterized by inflammatory infiltrates composed predominantly of lymphocytes and eosinophils. Many of these patients have anti-enterocyte antibodies,⁵³ specifically anti-harmonin, a 75-kDa scaffold protein of the gut epithelium (also known as Usher syndrome 1C protein),^{57,58} and anti-villin, a 95-kDa protein involved in the organization of actin cytoskeleton in the brush border of epithelial cells.^{59,60} Given the high frequency of detection of these antibodies in patients with IPEX syndrome with enteropathy, their presence was proposed as a diagnostic for IPEX syndrome.⁶¹ Antibodies against type I interferons capable of blocking interferon signaling can also be found, but their clinical relevance is not yet understood.⁶² Notably, anti-interferon antibodies in patients with IPEX syndrome are detected at lower levels and are less biologically potent than anti-interferon antibodies detected in patients with autoimmune polyendocrinopathy, candidiasis, ectodermal dystrophy syndrome (also known as autoimmune polyendocrine syndrome type 1, caused by a deleterious mutation of the *AIRE* gene),⁶² in which their presence is linked to resistance to T1D.⁶³ Antibodies against platelets, neutrophils, and liver and pancreatic islet proteins and others have also been described in relation to the target organs affected.⁵²

In patients with IPEX syndrome, most lymphocyte subsets have a normal distribution, with the exception of the amount of CD4⁺ T cells, eosinophils, and IgE, values of which are frequently found to be increased. Although the frequency of FOXP3⁺ Treg cells, when characterized by surface phenotype, varies from zero to comparable or even greater than in healthy control subjects,^{52,53,64-66} the amount of DNA demethylation at *FOXP3* TSDR is consistently higher in patients with IPEX syndrome, even at birth,²⁰ which is unique to this syndrome. Despite sometimes normal FOXP3 expression level, Treg cells of patients with IPEX syndrome are unable to suppress Teff cells.^{67,68} In addition, FOXP3⁺ Teff cells display a skewed phenotype toward TH2, TH17, and memory T cells; increased proliferation; and reduced TCR repertoire.^{69,70} Despite clear abnormalities demonstrated in the Teff cell compartment of patients with IPEX syndrome and in a FOXP3 knockout experimental models,⁶⁹ the current consensus is that the severe autoimmune phenotype in patients with IPEX syndrome results from impairment in the Treg cell compartment.⁷¹

The reason FOXP3 deficiency manifests predominantly as gut-, islet-, and skin-specific autoimmunity is still unclear. The fact that loss of FOXP3 function bestows severe pathology sometime in prenatal age^{72,73} points to an essential protective role of Treg cells during development of these organs. Alternatively, other regulatory cell subsets, such as TR1 cells, can compensate for the loss of Treg cell-mediated tolerance in other tissues.

A rare autosomal recessive deficiency in CD25, or the IL-2 receptor α chain, manifests as an IPEX-like syndrome.⁷⁴⁻⁷⁷ IL-2 is required for both the initiation and maintenance of adaptive T-cell responses and survival and function of FOXP3⁺ Treg cells.⁷⁸ Therefore, patients with CD25 deficiency present with both immunodeficiency and autoimmunity. Four described patients exhibited distinct clinical presentations, including infections encompassing cytomegalovirus- and *Pneumocystis* species-caused lung disease, *Staphylococcus aureus*- and *Pseudomonas aeruginosa*-caused cellulitis, candidiasis, and varicella infection. In addition to enteropathy, the most common disease manifestations are lymphadenopathy,

hepatosplenomegaly, and eczema. Other less common manifestations include T1D, autoimmune thyroiditis, autoimmune neutropenia, alopecia, and asthma.

In all patients described, surface CD25 expression was completely abrogated, allowing diagnosis based on immune phenotyping with flow cytometry. One patient had low total T-cell counts, and 3 of 4 patients displayed an approximately 1:1 (CD4/CD8 T-cell) ratio. This might be because CD8⁺ T cells respond better to IL-2 because of a high expression level of CD122 (IL-2 receptor β subunit) and therefore have an advantage over CD25-deficient CD4⁺ T cells. Indeed, proliferating granzyme B-positive CD8⁺ T cells have been described in skin infiltrates, but their accumulation was thought to be due to defective Treg cell regulation.⁷⁵ Three patients had reduced proliferative T-cell responses to nonspecific and/or antigen-specific stimuli, which could be rescued in some cases by high doses of IL-2 or IL-15.⁷⁴⁻⁷⁷ Also, 2 patients had normal counts of FOXP3⁺ Treg cells, and the third had low counts. Notably, the lack of surface CD25 renders the isolation of Treg cells for functional studies difficult; however, a recent study demonstrated impaired suppressive capacity in CD25-deficient Treg cells purified as CD4⁺TIGIT⁺CD127⁻ T cells.⁷⁹

CTLA-4 is a negative regulator of T-cell responses that shares similar structure with the other coreceptor family members CD28 and ICOS (inducible T-cell costimulator). Like CD28, CTLA-4 binds to the costimulatory surface receptors CD80 and CD86 expressed on APCs but possesses greater affinity.⁸⁰ On CTLA-4 binding, CD80 and CD86 are removed from the APC surface through transendocytosis,⁸¹ thus prohibiting their further interaction with CD28 and blocking the critical “second signal” to TCR stimulation that allows for T-cell activation. Although CTLA-4 is generally induced on T-cell activation, it is constitutively expressed at high levels on FOXP3⁺ Treg cells and plays an important role in Treg cell homeostasis and suppressive activity.⁸² It also fine tunes follicular helper T-cell differentiation, which depends on CD28 signaling, thus indirectly regulating humoral immunity.⁸² Therefore CTLA-4 deficiency leads to uncontrolled Teff cell responses and abnormal Treg cell function.

Loss of *Ctla4* in mice results in lethal early-onset systemic autoimmunity.^{83,84} CTLA-4 haploinsufficiency in human subjects leads to dominant immune dysregulation,^{85,86} which is also known as CTLA-4 haploinsufficiency with autoimmune infiltration (CHAI) disease⁸⁷ and presents with autoimmune cytopenia, hypogammaglobulinemia, infectious and noninfectious lung disease, enteropathy, lymphoproliferation, skin disease, and neurological involvement.^{87,88} Lymphoproliferation is mostly nonmalignant, presenting as lymphadenopathy, splenomegaly, and hepatomegaly, but multiple cases of associated lymphoma were reported, sometimes as a first symptom of the disease.⁸⁸ Infections in patients with CTLA-4 haploinsufficiency with autoimmune infiltration patients are caused by multiple pathogen species: common bacterial infections are caused by *Haemophilus influenzae*, *Streptococcus pneumoniae* (lung), and *Salmonella enteritidis* (gut); fungal infections are caused by *Candida* and *Aspergillus* species; and EBV and cytomegalovirus reactivation also occurs, affecting multiple organ systems.⁸⁸

In addition to hypogammaglobulinemia, patients with CTLA-4 haploinsufficiency often have decreased counts of total circulating B and memory B cells, natural killer (NK)

TABLE I. Clinical presentation, Treg cell phenotype, and functional assays used in patients with monogenic Tregopathies and IL-10/IL-10R deficiency

	Disease	Gene	Inheritance	Main clinical phenotype	Treg cell phenotype*	In vitro Treg cell function	Common or recommended functional validation of the mutation†	References
1	IPEX syndrome	<i>FOXP3</i>	XL	Enteropathy, T1D, eczema, failure to thrive, cytopenia, thyroiditis, hepatitis	Normal or ↓ circulating FOXP3 ⁺ Treg cells and FOXP3 expression level; high TSDR demethylation	Impairment in Treg cell suppression	FOXP3 expression in Treg cell and activated T cells; Treg cell-mediated suppression of Teff cell proliferation	67,68
2	CD25 deficiency	<i>IL2RA</i>	AR	Enteropathy, eczema, lymphoproliferation, recurrent infections	Normal or ↓ circulating FOXP3 ⁺ Treg cells	Impairment in Treg cell suppression (1 patient tested)	Surface CD25 protein expression by flow cytometry; STAT5b phosphorylation in response to IL-2; polyclonal and antigen-specific T-cell proliferation	74-77,79
3	CTLA-4 deficiency (ALPSV; CHAI)	<i>CTLA4</i>	AD	Enteropathy, T1D, cytopenia, respiratory disease	Normal or ↑ circulating FOXP3 ⁺ Treg cells and ↓ in FOXP3 expression level	Impairment in Treg cell suppression	Intracellular CTLA-4 protein level on memory Treg cells by flow cytometry at baseline and after stimulation; CD80-Ig uptake assay; CTLA-4 transendocytosis	81,85,86,95
4	LRBA deficiency (LATAIE)	<i>LRBA</i>	AR	Enteropathy, cytopenia, lymphoproliferation, recurrent infections	Normal or ↓ circulating FOXP3 ⁺ Treg cells and FOXP3 expression level	Impairment in Treg cell suppression	Intracellular CTLA-4 protein level on memory Treg cells by flow cytometry at baseline and after stimulation; LRBA expression on PHA-stimulated PBMCs by flow cytometry	89,91,95,96
5	BACH2 deficiency	<i>BACH2</i>	AD	Enteropathy, lymphoproliferation, recurrent respiratory tract infections	↓ in FOXP3 expression level on Treg cells	Undetermined	BACH2 protein level by flow cytometry and <i>PRDM1</i> mRNA level by PCR	97
6	STAT3 GOF (ADMIO, ADMIO1)	<i>STAT3</i>	AD	Enteropathy, cytopenia, T1D, lymphoproliferation, recurrent infections	Normal or ↓ circulating FOXP3 ⁺ Treg cells	Variable impairment in Treg cell suppression	STAT3 response to IL-6 stimulation (eg, through overexpression of WT or mutant gene into HEK293 cells containing STAT3-responsive luciferase)	98-100
7	IL-10 or IL-10R deficiency	<i>IL10</i> , <i>IL10RA</i> , <i>ILRB</i>	AR	Enterocolitis, perianal disease, folliculitis, infections	Treg cells or T _R 1 cells undetermined	Undetermined	T _R 1 cells: CD49b/LAG3 staining by flow cytometry; IL-10 deficiency: suppression of TNF-α production by synthesized mutant IL-10; IL-10R deficiency: STAT3 response to IL-10 stimulation	36,101

The "References" column shows a nonexhaustive list of studies using indicated functional tests.

↓, Decrease in frequency and/or absolute count and/or expression level per cell.

AD, Autosomal dominant; ADMIO, autoimmune disease, multisystem, infantile-onset syndrome (also known as STAT3 GOF disease, caused by a GOF mutation in the *STAT3* gene); ALPSV, autoimmune lymphoproliferative syndrome, type V; AR, autosomal recessive inheritance; CHAI, CTLA-4 haploinsufficiency with autoimmune infiltration; LATAIE, LRBA deficiency with autoantibodies, Treg cell defects, autoimmune infiltration, and enteropathy; XL, X-linked.

*Treg cell phenotype was not uniformly defined across these studies; for simplicity, this implies FOXP3⁺CD4⁺CD3⁺ T cells with variable addition of other markers, such as CD25 and CD127.

†All these assays are carried out in parallel in patients and healthy subjects.

cells, and T cells and occasionally increased frequencies of CD4⁺ helper T cells.⁸⁸ Circulating Treg cells have been reported both at normal and increased frequency,^{85,88} but Treg cell phenotyping can be confounded by lower expression of CD25, which is a surface marker used to define Treg cells.⁸⁵ Indeed, when measured based on TSDR demethylation, both the Treg cell compartment and the Treg cell/Teff cell ratio are decreased (our unpublished observations). As in patients with IPEX syndrome, CTLA-4-deficient Treg cells also display decreased suppressive capacity.^{85,86}

Homozygous or compound heterozygous deficiency in *LRBA* causes secondary CTLA-4 deficiency, also known as LRBA deficiency with autoantibodies, Treg cell defects, autoimmune infiltration, and enteropathy (LATAI) disease, which shares phenotypical and clinical features with CTLA-4 haploinsufficiency.⁸⁹ CTLA-4 continuously cycles to and from the cell surface, but the majority of the protein is located intracellularly. LRBA mediates CTLA-4 recycling from the endosomes to the T-cell surface to prevent lysosomal degradation. Accordingly, LRBA-deficient Treg cells display decreased CTLA-4 expression levels. Similar to patients with CTLA-4 mutation, LRBA-deficient T cells are hyperproliferative on *in vitro* activation, and Treg cell-mediated suppression and CD80 transendocytosis are impaired.^{90,91} However, LRBA deficiency often starts in young children, whereas CTLA-4 haploinsufficiency affects older children or young adults.⁸⁷ LRBA deficiency also displays almost complete penetrance, whereas the recent analysis of 133 *CTLA4* mutation carriers revealed approximately 67% penetrance.⁸⁸

The disease-causing *LRBA* mutation was identified during genetic screening of patients with combined variable immunodeficiency,⁹² IPEX syndrome-like disease,⁹¹ and neonatal T1D.⁹³ Immunologically, patients had normal or reduced lymphocyte counts, mostly normal T-cell levels with reduced Treg cell counts, normal or low total B-cell counts with a reduced memory compartment,^{89,92} and increased numbers of circulating follicular helper T cells.^{91,94} Infections, especially in the lung, were common and caused by various bacterial, viral, and fungal microorganisms.⁸⁹

Distinguishing and confirming functional consequences of CTLA-4 and LRBA mutations, which similarly affect CTLA-4 baseline expression, requires *in vitro* approaches, such as measuring CTLA-4 induction after stimulation⁹⁵ or LRBA intracellular staining using flow cytometry (Table I).^{*} Moreover, differences in onset and clinical presentation of these 2 diseases might be due to the yet unknown LRBA functions in B cells, which display defective autophagy, activation, and survival in LRBA-deficient patients.⁹²

BACH2 is a basic region-leucine zipper transcription factor required for B cell class-switch recombination and somatic hypermutation, T-cell differentiation and function, and alveolar macrophage function. The *BACH2* locus contains one of the most prominent T-cell superenhancers,¹⁰² which control genes important for cell identity.¹⁰³ Because genetic variation in superenhancers is commonly associated with disease risk, single nucleotide polymorphisms in the *BACH2* region are linked to multiple autoimmune and inflammatory conditions: rheumatoid arthritis, T1D, asthma, multiple sclerosis, vitiligo, Graves disease, Crohn disease, and celiac disease.¹⁰⁴⁻¹⁰⁶ Disruption of

Bach2 expression in mice results in inefficient Treg cell generation accompanied by Teff cell expansion and tissue infiltration.¹⁰⁷ BACH2 competes with the activator protein 1 family of transcription factors to passively repress T-cell signaling pathways downstream of the TCR.¹⁰⁸ Also, it seems to be required for regulation of T_H2 cytokine production and T_H2-mediated airway inflammation.^{109,110} In Treg cells BACH2 stabilizes their phenotype, promotes their survival, controls their activation, and regulates the high expression level of FOXP3.¹¹¹

BACH2 haploinsufficiency was described last year in 3 patients from 2 families.⁹⁷ The clinical phenotype consisted mainly of lymphadenopathy, chronic diarrhea, IBD, and recurrent sinopulmonary infections. Both mutations resulted in decreased BACH2 protein stability and lower BACH2 protein expression in B and T cells despite normal *BACH2* mRNA levels. Levels of *PRDM1* (*BLIMP1*) mRNA, a gene expressed by BACH2, were increased. Patients had a uniform decrease in the frequency of total memory and IgG-switched memory B cells, and 2 of 3 patients had low circulating IgM, IgG, IgA, and IgE levels. All 3 patients had decreased levels of FOXP3⁺ Treg cells in the blood compared with healthy control subjects and control subjects with IBD; in 1 patient FOXP3⁺ Treg cells were also decreased in the colon. All 3 patients had increased levels of T-bet, and 2 of 3 patients displayed decreased CD4⁺ T-cell proliferation in response to TCR-mediated stimulation. It is yet unclear whether decreased proliferation is due to the intrinsic defect in patients' CD4⁺ T cells or from increased apoptosis of activated cells, as observed in *Bach2*-deficient mice.^{108,109}

STAT3 is a transcription factor activated downstream of many cytokine receptor families, including glycoprotein 130 (gp130; with members IL-6 and IL-27), IL-10, IL-23, several common γ chain cytokines, interferons, and others. Cytokine receptor triggering turns on the receptor-associated Janus-activating kinases (JAKs), which phosphorylate the tyrosine residues on the receptor, creating SH2-binding sites. Docking of STAT proteins to these sites leads to their phosphorylation, after which they dissociate, create homodimers or heterodimers, and traverse to the nucleus to initiate the transcription of many target genes.¹¹²

LOF mutations in *STAT3* result in hyper-IgE syndrome (also known as Job syndrome or Buckley syndrome; the autosomal dominant form is caused by a deleterious LOF mutation of the *STAT3* gene, and autosomal recessive hyper-IgE syndrome also exists and is caused by mutations in *DOCK8* and several other genes).^{113,114} GOF mutations in *STAT3* were described more recently after a genomic screen in patients with early-onset polyautoimmunity (T1D, autoimmune hypothyroidism, and celiac disease).⁹⁸ Discovery of more than 20 new patients followed shortly thereafter.^{99,100,115-119} Autoimmune disease, multisystem, infantile-onset syndrome (ADMIO, also known as *STAT3* GOF disease, which is caused by a GOF mutation in *STAT3* gene) usually manifests early in infancy or childhood and is presented commonly with diarrhea, enteropathy, autoimmune cytopenias, lymphoproliferation, interstitial lung disease, recurrent infections with various pathogen species, eczema, short stature, and T1D; while less commonly with hypothyroidism, arthritis, hepatitis, and malignancies (Hodgkin lymphoma, T-cell large granular lymphocyte leukemia, and squamous cell carcinoma). Germline GOF *STAT3* mutations are found across the entire gene and do not seem to correlate with the patient phenotype.^{100,112} Interestingly, the same missense mutation can

*References 36,67,68,74-77,79,81,85,86,89,91, and 95-101.

lead to both GOF or LOF, depending on the electrostatic interaction with DNA that affects binding affinity.^{98,120} Unlike in patients with STAT1 GOF disease, some GOF mutations in *STAT3* did not affect its baseline phosphorylation but increased its transcriptional activity.¹⁰⁰

Dysregulation in B-cell subsets and immunoglobulin levels was frequently observed but highly variable; several patients exhibited lower levels of class-switched immunoglobulin (IgG and IgA) and numbers of memory B cells,^{99,100,115–118} indicating a potential impairment in B-cell antigen responses. Many, but not all, patients had decreased frequencies of FOXP3⁺ Treg cells, and loss of suppressive function has been reported in some patients.^{98,99,116,117} Lymphoproliferation was accompanied by an increase in the frequency of circulating double-negative CD3⁺CD8[−]CD4[−] T cells and defective Fas-mediated apoptosis.^{100,117} Several patients had low T_H17 numbers,^{99,115} and others had normal or even increased T_H17 responses.¹¹⁹

The STAT3 signaling cascade involves multiple proteins that regulate Treg and T_H17 cells. STAT3 is strongly induced by IL-6, which, together with TGF- β , is the key cytokine required for T_H17 differentiation.¹²¹ IL-6 inhibits TGF- β -induced FOXP3 upregulation in naive CD4⁺ T cells, polarizing them from Treg cells toward a potentially pathogenic T_H17 phenotype.¹²² In addition, STAT3 activation upregulates the expression of secretion of cytokine signaling 3 (SOCS3), which inhibits STAT5, a positive regulator of FOXP3 and CD25 expression. An overexpression of SOCS3 could thus impair Treg cell responses.¹²³ Indeed, increased SOCS3 and impaired STAT5 phosphorylation were found in STAT3 GOF cell lines,¹⁰⁰ and patients with STAT3 GOF mutations share some clinical features with STAT5b-deficient patients.¹²⁴ STAT5b also inhibits IL-17 transcription, whereas STAT3 has the opposing effect.^{125,126} SOCS3 is a negative regulator of IL-17 production and T_H17 generation.^{127,128} Therefore in certain patients with STAT3 GOF mutations, the inhibitory effect of SOCS3 on T_H17 responses may overcome T_H17-promoting signals delivered through the increased function of STAT3 and suppression of STAT5. It is likely that autoimmune manifestations in patients with STAT3 GOF disease result both from the impaired FOXP3 upregulation and imbalanced Treg/T_H17 cell polarization axis, but it is unclear which has the dominant effect. Because of the broad role of STAT3 in governing signaling cascades, differentiation, growth, and regeneration in many tissues and cell types, other cell subsets and pathways can contribute to immune dysregulation in patients with STAT3 GOF disease.

Other monogenic immune diseases that affect Treg cell number or function include deficiencies in STAT5b, ITCH (itchy E3 ubiquitin protein ligase), ITK (IL-2 inducible T cell kinase; reviewed recently here¹²⁹), and nuclear factor κ B subunit 2 (NFKB2).¹³⁰ However, in the IUIS 2017 classification, these diseases were categorized separately from the IUIS-defined Treg cell diseases reviewed above.¹ Autosomal recessive STAT5b deficiency is categorized as “combined immunodeficiency with associated or syndromic features” because the unifying symptom in 9 patients described thus far is severe growth delay insensitive to the growth hormone (GH).^{131–140} Namely, STAT5b deficiency impairs GH receptor signal transduction, manifesting with normal circulating GH levels, low insulin-like growth factor 1 levels, and nonresponsiveness to GH therapy. Prolactin levels are high in most patients. Consistent with other cases of GH insensitivity,

STAT5-deficient patients commonly have mild dysmorphic features and delayed puberty. Unlike in patients with GH insensitivity, all but 1 STAT5-deficient patient also presented with immune dysregulation, most commonly interstitial lung disease, eczema, recurrent skin and respiratory tract infections, and various other autoimmune manifestations.¹⁴¹ Lung disease was often severe, sometimes with lethal outcomes¹³² or requiring lung transplantation.^{137,138} Immune dysregulation in STAT5b-deficient patients is attributable to STAT5b's crucial role in the IL-2 receptor signaling cascade.¹²⁴ IL-2 supports Teff cell proliferation and survival, maintenance of FOXP3⁺ Treg cells and is essential for sustaining FOXP3 expression. Indeed, most STAT5b-deficient patients have decreased numbers of circulating T cells with a preserved CD4/CD8 ratio and in some cases decreased T-cell proliferation in response to non-specific and antigen-specific stimulation.¹⁴¹ When measured, the frequency of CD25^{high}CD4⁺ Treg cells in STAT5b-deficient patients was decreased,^{131,133} FOXP3 expression levels in CD25^{high} Treg cells were very low, CD4⁺ T-cell apoptosis was increased, and patients' CD25^{high} Treg cells did not suppress responder T-cell proliferation.¹³³ Some STAT5-deficient patients also had mild lymphopenia, with decreased peripheral $\gamma\delta$ T cells, B cells and/or NK cells, and hypergammaglobulinemia.

Autosomal dominant STAT1 GOF mutations are classified by IUIS as “Defects in intrinsic and innate immunity: Predisposition to mucocutaneous candidiasis.” The STAT1 GOF mutation primarily impairs development of IL-17–producing cells. The disease presents most often as chronic mucocutaneous candidiasis and carries susceptibility to other fungal and bacterial infections, although a subset of patients have demonstrated autoimmunity and IPEX-like syndrome despite wild-type FOXP3.^{142,143} In 184 documented cases of heterozygous STAT1 GOF disease, *STAT1* mutations disrupted the coil-coil or DNA-binding domains. This resulted in impaired STAT1 dephosphorylation after upstream receptor triggering by cytokines, such as IFN- α and IFN- γ .

Because of the more prominent effect on T_H17 cells, only limited data exist on the effects of STAT1 GOF disease on the Treg cell compartment. In a 2013 study of patients with chronic mucocutaneous candidiasis showing IPEX-like features, 5 identified heterozygous patients with STAT1 GOF disease had normal percentages of FOXP3⁺ Treg cells compared with healthy control subjects.¹⁴⁴ Subsequent functional testing of Treg cells from 2 of the patients also showed no observable defects. However, a 2016 case study described a single patient with STAT1 GOF disease with decreased FOXP3⁺ Treg cell frequency, which correlates with our unpublished observations. Thus it remains unclear whether STAT1 GOF mutation-mediated IPEX-like syndrome is driven by a defect in Treg cells.¹⁴⁵ The autoimmunity observed in these patients can also be driven by prolonged STAT1 signaling after physiologic type I interferon receptor triggering, as observed in *bona fide* interferonopathies (ie, inborn errors of immunity caused by a genetic mutation that leads to an upregulation of type I interferon signaling), another class of monogenic diseases associated with autoimmunity and autoinflammation.¹⁴⁶

ITCH deficiency, another monogenic disease associated with Treg cell abnormalities, is classified by IUIS as “Disease of immune dysregulation: Autoimmunity with or without lymphoproliferation”; ITK deficiency is classified as “Disease of immune dysregulation: Susceptibility to EBV and lymphoproliferative

conditions”; and NFKB2 deficiency is classified as “Predominantly antibody deficiency.” Importantly, as for STAT5b deficiency and STAT1 GOF, the IUIS classification does not preclude Treg cell defects in patients with these conditions. Instead, the predominant disease presentation is different than that in diseases primarily associated with Treg cell insufficiency. Similarly, there are many more monogenic immune diseases in the IUIS 2017 classification that also present with autoimmunity, although a direct involvement of Treg cells in the pathogenesis remains to be defined.

IL10 AND IL10R DEFICIENCY: A PUTATIVE DEFECT OF T_R1 CELLS?

IL-10 plays a crucial role in maintaining immune homeostasis and preventing aberrant and excessive inflammation.¹⁴⁷ In addition to T_R1 cells, IL-10 is produced by several hematopoietic cell subsets, such as other T cells, Treg cells, monocytes, macrophages, dendritic cells, and B cells. IL-10 receptor (IL-10R) is a heterotetramer consisting of 2 IL-10R α and 2 IL-10R β subunits that are encoded on the *IL10R1* and *IL10R2* genes, respectively.^{148,149} The activated IL-10R signaling complex phosphorylates STAT3 through JAK1 and tyrosine kinase 2 (TYK). Phospho-STAT3 homodimers are responsible for activating the IL-10–responsive anti-inflammatory response, resulting in suppression of proinflammatory pathways acting through IL-6 and Toll-like receptor 4.¹⁵⁰ The anti-inflammatory role of IL-10 is particularly important within the *in vivo* environment of the gut, which is constantly exposed to potentially immunogenic stimuli through the presence of commensal microbiota and foreign food antigens.¹⁴⁸

Deficiencies in *IL10* or either of the IL-10R subunits (collectively *IL10R*) are autosomal recessive inherited disorders that cause hyperinflammation because of nonfunctional IL-10 signaling. The IUIS categorizes *IL10* and *IL10R* mutations as “Diseases of immune dysregulation: Immune dysregulation with colitis.”¹⁵¹ Underscoring the importance of IL-10 in maintaining immune homeostasis in the gut, IL-10 deficiency was first described in preclinical models in which *Il10*^{−/−} mice developed spontaneous enterocolitis that could be resolved by means of administration of recombinant IL-10.^{152,153} Likewise in human subjects, *IL10/IL10R* deficiency has been shown to cause a monogenic form of very early-onset (veo) IBD that is refractory to pharmacologic immunosuppressive regimens (corticosteroids, methotrexate, and anti-TNF- α).¹⁵⁴ Mutations in *IL10* and *IL10R* subunits causing veo-IBD are rare, with approximately 60 cases described in the literature,¹⁵⁴ of which only 5 have a mutation in *IL10*.^{154–156} Patients with defective IL-10 signaling commonly present with bloody diarrhea, perianal disease, and enterocolitis but otherwise show only minor abnormalities in immune cell populations or serum immunoglobulin levels.¹⁰¹ Patients with mutations in *IL10R2* might show increased disease severity or extraintestinal symptoms, such as folliculitis. This may be caused by the additional impairment of other IL-10 family cytokines because IL-10R2 is also utilized by IL-22, IL-26, IL-28, and IL-29 receptors.¹⁰¹ Patients with *IL10R* mutations were also reported to have B-cell lymphoma.¹⁵⁷

In addition to genetic testing for mutation, functional testing can be used to reveal the nature of the *IL10/IL10R* mutation.¹⁰¹ PBMCs isolated from patients with *IL10R* deficiencies show neither phosphorylation of STAT3 nor a decrease in LPS-induced TNF- α expression when treated with recombinant

Mutated genes that affect Treg function:



*Except STAT3, all mutations result in the loss of function

Main symptoms (common to most diseases):

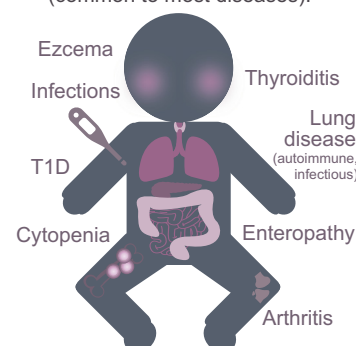


FIG 2. Tregopathies: graphic summary of mutated genes and main symptoms. For clarity, only the diseases defined by IUIS as Treg cell defects are included.

IL-10. PBMCs isolated from patients with IL-10 defects respond normally to *in vitro* IL-10 stimulation; thus *in vitro* synthesis and testing of the mutant IL-10 is required to confirm loss of function. Because patients with IL-10/IL-10R deficiency tend to be refractory to standard immunosuppressive therapy, the only curative treatment at the moment is HSCT.¹⁵⁸ Although long-term patient monitoring data are not yet available because the first transplantation was only described in 2009, 16 patients have successfully received HSCT with complete remission of colitis.^{154,156,158–161}

A recent study of 12 patients with IL-10R deficiency found normal FOXP3 expression levels and normal frequencies of FOXP3⁺ Treg cells in patients’ blood and intestinal mucosa, and patients’ FOXP3⁺ Treg cells exhibited comparable suppressive capacity to healthy control Treg cells.¹⁶² On the other hand, the crucial role of IL-10 for T_R1 function strongly supports the hypothesis that loss of T_R1 function contributes to the disease phenotype in patients with *IL10/IL10R* deficiency. In SCID mice adoptive transfer of *in vitro*–generated antigen-specific T_R1 cells prevented colitis induced by pathogenic T cells,³⁷ whereas coadministration of wild-type T_R1 cells could prevent colitis in another murine IBD model induced by transferred CD45RB^{high} or T_H17 T cells.¹⁶³ These data from animal models are also supported by the aforementioned study of intestinal T_R1 cells in adults with IBD.⁵¹ However, T_R1 phenotyping has not been conducted thus far in patients with *IL10/IL10R* deficiency.

DIAGNOSIS OF MONOGENIC DISEASES THAT AFFECT TREG CELL FUNCTION: GENETIC, EPIGENETIC, PHENOTYPIC, AND FUNCTIONAL ASSESSMENTS

Despite the growing genetic diversity, it has become clear in the past decade that clinical manifestations of Tregopathies (Fig 2 and Table I) are dominated by noninfectious gut pathology, cytopenias, endocrine abnormalities (especially T1D), and lung and skin inflammation. Also, liver, kidney, and neurologic disorders are increasingly reported, but the number of patients for many diseases is still too small to clearly cluster and associate discrete clinical manifestations to a specific gene defect. When mutations also affect the function of effector immune cells,

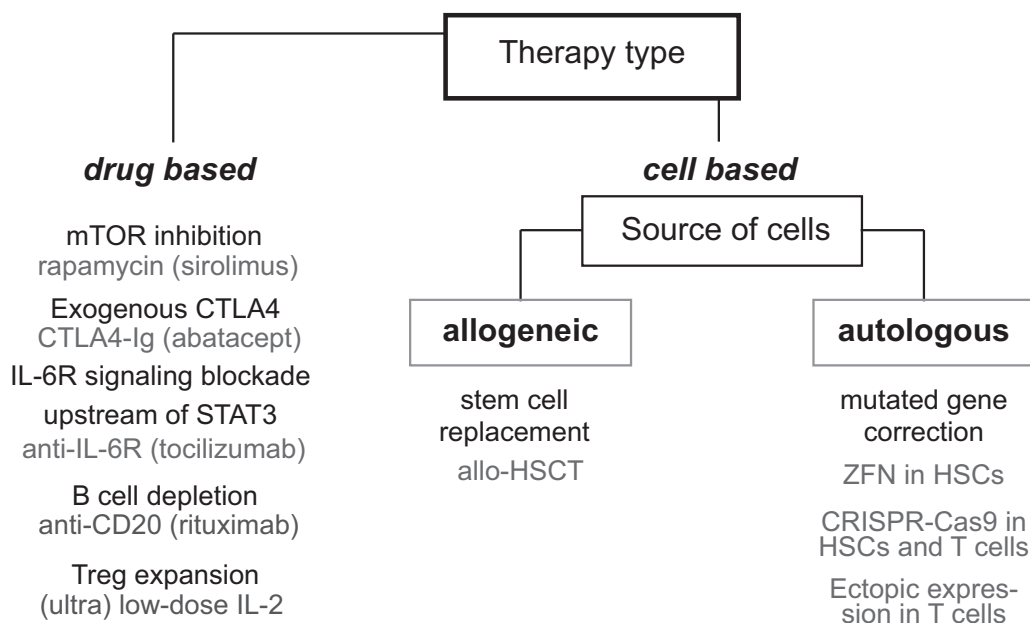


FIG 3. Therapeutic options in patients with Tregopathies. For drug-based therapeutic options, the most common antibodies and targeted small-molecule immune modulators are displayed. Nonspecific immunosuppressive agents used in alleviating autoimmune inflammation, such as corticosteroids, cyclosporine, and others, are not listed. Cell-based therapies are divided according to the cell source: patient-derived (autologous) or healthy donor (allogeneic). *IL-6R*, IL-6 receptor; *mTOR*, mammalian target of rapamycin; *ZFN*, zinc finger nuclease.

infections are often part of the phenotype. Notably, infections can be found together with nonmalignant lymphoproliferation rather than lymphopenia. Typically, onset of these symptoms early in childhood, accompanied by their persistency, recurrence, and/or poor therapeutic responses to standard first-line treatments, should prompt further diagnostic investigations into the possible genetic origin of the disease. The timely diagnosis of patients with later-onset and/or overlapping autoinflammatory manifestations can be more challenging. Diagnosis is further complicated by heterogeneity of clinical presentations because of different penetrance of certain gene mutations between carriers or the effect of individual mutation site on protein function.

A helpful tool for diagnosis of Tregopathies is the IUIS 2017 update on phenotypic classification for 354 known inborn errors of immunity, containing basic algorithms for all primary immunodeficiency categories and accompanied by a smartphone app.^{151,164} Ultimately, however, only a systematic prospective multicenter study will elucidate the characteristics of each disease's onset and dissect the diversity of presentations and evolution.

A prerequisite for the confirmed diagnosis of Tregopathies is the detection of a mutation predicted to be deleterious and present in less than 1% of the population in patients' DNA by using whole-genome, whole-exome, or targeted DNA sequencing.¹⁶⁵ Sequencing usually follows extensive clinical workup and laboratory testing of patients presenting with 1 or more of the typically described autoimmune manifestations (Fig 2), especially when early severe onset or poor response to therapy is observed.

Because of (1) the great overlap in clinical presentations between individual disorders, (2) the accelerated discovery of

new inborn errors of immunity in the era of next-generation sequencing,¹ and (3) the possible large underestimate in the prevalence of these diseases,¹⁶⁶ functional confirmation of newly discovered mutations is recommended.¹⁶⁷ Frequently, these additional assays are not provided by certified clinical diagnostic laboratories, and functional confirmation depends on performing assays in research laboratories associated with attending physicians or through collaborative efforts with other researchers. For example, although the assessment of Treg cell frequency by using flow cytometry is offered as a clinical laboratory-certified assay in several laboratories in the United States, interpretation of the assay results in patients with active autoimmune disease can be challenging because *in vivo*-activated Teff cells express some of the Treg cell-associated markers.^{66,68,168,169} In addition, Treg cells can be dysfunctional, even when present in normal numbers, as in patients with IPEX syndrome. Confirming the functional effect of a FOXP3 mutation in patients with IPEX syndrome thus requires an *in vitro* suppression assay with purified Treg and Teff cells,¹⁷⁰ which is thus far only performed in research laboratories.

Furthermore, using standard FOXP3 staining by means of flow cytometry to define Treg cells (in addition to other markers, such as CD3, CD4, CD25, and CD127) can be misleading in patients with inflammatory conditions in which activated Teff cells are present *in vivo*. More accurate results can be obtained by determining the TSDR demethylation status, which distinguishes *bona fide* FOXP3⁺ Treg cells with stable suppressive function from activated FOXP3⁺ Teff cells or *in vitro*-induced Treg cells with transient suppressive capacity.^{171,172} We used the TSDR demethylation analysis in 28 patients with IPEX-like syndrome to demonstrate that Treg cell counts were

significantly decreased, despite the variable frequency of FOXP3⁺ Treg cells by using flow cytometry.¹⁴ Epigenetic quantitative PCR to detect lineage-specific demethylated regions for CD4⁺ and CD8⁺ T cells, Treg cells, B cells, NK cells, monocytes, and neutrophils has been recently developed and proposed as a potential immune diagnostic test for newborn diagnosis of primary immunodeficiencies, including IPEX syndrome.²⁰

The mutations, main clinical presentations, Treg cell phenotype and function, and functional experiments to confirm the mutation effect in Tregopathies, along with accompanying references, are summarized in Table I. Table I also includes IL-10/IL-10R deficiency; functional assays used to differentiate between IL-10 and IL-10R defects are reviewed here.¹⁰¹

TREATMENT APPROACHES TO ALLEVIATE FOXP3⁺ TREG CELL DEFECTS

Treatment of monogenic diseases can either be curative or aim at modulating the pathogenic effects of the disease to provide clinical benefit. Treatment of Treg cell deficiencies can be challenging because it often requires both immunosuppression and protection against infections. Nonspecific immunosuppressive drugs, such as corticosteroids and calcineurin inhibitors, are still widely used, often to control the inflammation and T-cell activation before diagnosis is established. Treatment with biologics (biopharmaceuticals; ie, drugs derived from biological sources, such as recombinant antibodies or cytokines) might be more efficient in controlling inflammation, but their effects might also not be specific. In addition, even when efficacious, biological agent-based therapy cannot correct the mutated gene. For diseases that affect the immune system, cure can be achieved either by replacing hematopoietic stem cells (HSCs) by means of transplantation of allogeneic stem cells from healthy donors or by correcting the mutated gene in a patient's own stem cells. Below, we will discuss some of the commonly used immune-modulating and curative treatments for monogenic diseases that affect Treg cell function (Fig 3).

Rapamycin (sirolimus) is a small molecule that inhibits the mTOR pathway. Rapamycin is used extensively as a nonspecific immunosuppressive drug to prevent graft rejection or graft-versus-host disease after transplantation.^{173,174} Because of its selective inhibitory effect on T_H17 cell proliferation while sparing rapamycin-resistant Treg cells, this immunosuppressive drug increases the Treg/T_H17 cell ratio, supporting the relative expansion and function of Treg cells.^{175,176} Therefore rapamycin treatment can be beneficial in immune-regulatory disorders. Recently, we have shown that rapamycin has significant clinical benefits in patients with IPEX syndrome, despite the Treg cell dysfunction caused by FOXP3 mutations.^{52,53}

CTLA4-Ig (abatacept) is a fusion protein consisting of an IgG₁ Fc domain fused to the CTLA-4 extracellular domain. It has been successfully used to control autoimmune inflammation in patients with CTLA-4 haploinsufficiency^{88,177} and also in patients with LRBA deficiency, with improvement in overall clinical conditions and interstitial lung disease and few side effects.⁹⁰ The majority of CTLA-4-deficient patients also had a good clinical response to rapamycin.⁸⁸ In addition, hydroxychloroquine has been suggested as a potential treatment for LRBA deficiency because of its inhibitory effect on lysosomal degradation.^{89,90}

The mAb against the IL-6 receptor (tocilizumab) has been tested to treat STAT3 GOF disease.¹⁰⁰ IL-6 is a proinflammatory cytokine that activates the STAT3 signaling cascade and supports T_H17 differentiation. Tocilizumab therapy led to improvement of joint contractures by shrinking the expanded T_H17 population and increasing numbers of circulating Treg cells in patients with STAT3 GOF disease.^{100,116}

The mAb against CD20 (rituximab) primarily modulates humoral immune responses by depleting B cells. Its use in immune-regulatory disorders aims to reduce the pool of autoantibody-secreting cell precursors. In addition, B cells can also present antigen, although less efficiently than professional APCs.¹⁷⁸ Rituximab is used in autoimmune cytopenias, which is a very common symptom in diseases of immune dysregulation,^{179,180} and in other conditions such as granulomatous lymphocytic interstitial lung disease.¹⁸¹ In patients with immunoregulatory disorders, rituximab has been administered thus far to patients with *FOXP3*, *CD25*, *CTLA4*, *LRBA*, and *STAT3* GOF mutations in combination with other drugs.^{53,74,88,89,100,182} Rituximab has not been tested yet in patients with *BACH2* deficiency, but it might not necessarily be beneficial because of already low memory B-cell counts and decreased immunoglobulin levels in this syndrome.⁹⁷ Indeed, in children with systemic lupus erythematosus (SLE)-associated cytopenia treated with rituximab, 3 of 4 patients with pre-existing low IgG levels developed persistent hypogammaglobulinemia after rituximab that required immunoglobulin replacement.¹⁸³

The efficacy of rituximab to control cytopenia in patients with Tregopathies is currently difficult to establish, first because of the small number of patients in some of these diseases and second because other therapies frequently coadministered in patients with Tregopathies (steroids, rapamycin, or calcineurin inhibitors) can also contribute to the beneficial effect of rituximab.^{184,185} The abovementioned study on pediatric SLE demonstrated complete response in 23 of 24 children with cytopenia treated with rituximab.¹⁸³ In an adult SLE study, of 71 patients treated with rituximab for cytopenia, approximately 60% displayed a complete response, and relapses were successfully treated in most with a second round of rituximab therapy.¹⁸⁶ Similarly, in patients with combined variable immunodeficiency-associated cytopenia, the complete response rate was 74%, and patients who relapsed also mostly responded to rituximab readministration.¹⁸⁰ Importantly, rituximab-induced B-cell depletion increases the risk of infection, which needs to be considered while designing a therapeutic strategy for patients with Tregopathies, especially for diseases already associated with increased susceptibility to infection.

IL-2 is a potent T-cell growth factor and a critical cytokine for maintenance and function of FOXP3⁺ Treg cells, which constitutively express high levels of CD25, the high-affinity IL-2 receptor α subunit.¹⁸⁷⁻¹⁸⁹ High CD25 expression confers to Treg cells the ability to respond to low doses of IL-2, whereas T_H17 cells require higher IL-2 concentrations to support their proliferation. In the attempt to selectively expand Treg cells and promote immune regulation, administration of low-dose recombinant human IL-2 (aldesleukin) was tested to prevent graft-versus-host disease after HSCT, resulting in an increase in the numbers of circulating Treg cells.¹⁹⁰ IL-2 was also administered at various concentrations to patients with Wiskott-Aldrich syndrome, who have dysfunctional TCR signal

transduction, immunologic synapse formation, and Treg cell insufficiency. Five of 6 patients with Wiskott-Aldrich syndrome exhibited an increase in numbers of circulating Treg cells after therapy. However, increased platelet counts were found only after low-dose IL-2 (0.5 million units/m²/day); at 1 million units/m²/day IL-2, several patients experienced worsening of their thrombocytopenia.¹⁹¹ With such a narrow dynamic range, recombinant IL-2 might not become a widely used therapeutic strategy. Efforts are underway to modify the native IL-2 and use it concurrently with the administration of autologous engineered Treg cells expressing the receptor for modified but not native IL-2,¹⁹² thus avoiding the endogenous Teff cell activation caused by therapeutic IL-2 administration.

Cell and gene therapy strategies for monogenic diseases that affect Treg cell function aim to replace and/or genetically correct the relevant cell types. Replacing mutated hematopoietic cells with normal cells through allo-HSCT has been the only curative option for many monogenic immunoregulatory disorders. However, allo-HSCT carries a substantial risk for serious complications and mortality ascribed to the procedure itself, primarily because of graft-versus-host disease and the induced immunodeficient state before engraftment that impairs host defenses against pathogens. Pretransplantation control of inflammation, organ damage, and ablation of autoreactive Teff cells might decrease the risk of HSCT in diseases of immune dysregulation.¹⁹³

To evaluate the risks and benefits of allo-HSCT versus various immunomodulatory and immunosuppressive therapeutic regimens in patients with IPEX syndrome, we performed an international, multicenter, retrospective analysis of outcomes in 96 patients with IPEX syndrome.⁵³ The overall survival after HSCT versus pharmacologic or biological therapy was 73.2% and 65.1%, respectively, but HSCT was the only treatment conferring disease-free survival.

Allo-HSCT is also commonly applied in other monogenic immune diseases, and it is often the only life-saving option for severe cytopenias.⁸⁸ A recent study describing 133 *CTLA4* mutation carriers reported outcomes of 12 patients treated with allo-HSCT, as indicated by therapy-resistant cytopenias, enteropathy, or Hodgkin lymphoma. Of 3 deaths, 2 patients died from allo-HSCT complications and 1 from diabetic ketoacidosis after a period of full remission.⁸⁸ Allo-HSCT outcomes of patients with LRBA deficiency were recently summarized as well, reporting approximately 50% disease-free survival in 12 patients undergoing transplantation, 3 of whom achieved complete remission, 5 of whom achieved partial remission, and 4 of whom who died from HSCT-related complications.¹⁹⁴ Allo-HSCT has also been used in patients with CD25 deficiency¹⁹⁵ and STAT3 GOF disease,¹⁰⁰ but very low patient numbers make it difficult to summarize the outcomes. Allo-HSCT might not be the optimal treatment option for STAT5b deficiency because it is not expected to correct the growth abnormality. Overall, the high morbidity and mortality risk of allo-HSCT and the sometimes limited availability of appropriate donors places a strong emphasis on strategies aimed at correcting the mutated gene in a patient's own hematopoietic cells.

Monogenic diseases that affect Treg cell function are candidates for gene therapy because the main clinical effect of the mutation is on readily accessible immune cells. Correcting the mutated gene in HSCs should therefore lead to resolution of the

majority of disease manifestations. The gene transfer space has had more than 25 years of preclinical, clinical, and follow-up data showing the feasibility, safety, and efficacy of gene therapy for SCIDs.^{196,197} Strimvelis recently became the first approved treatment using autologous *ex vivo* HSC gene therapy for SCID caused by the adenosine deaminase mutation.¹⁹⁸ Historical approaches to gene correction of HSCs has taught us that the safety of gene therapy could be vastly improved by lentiviral optimization¹⁹⁹ and directing gene integration to a specific genomic site to prevent activation of potentially oncogenic genes.^{200,201} With clinical-grade lentiviruses, conventional CD4⁺ T cells can be reprogrammed through ectopic expression of FOXP3 to confer suppressive properties. We have tested this approach in preclinical studies for patients with IPEX syndrome, where CD4⁺ T cells derived from patients with IPEX syndrome that were transduced with wild-type FOXP3 exhibited a comparable phenotype and suppressive activity to healthy donor-derived Treg cells.^{70,202}

Additionally, regulation of gene expression for many genes identified in patients with Treg cell deficiencies is cell type specific. Therefore previous strategies that used constitutive active promoters^{203,204} in progenitor cells are not always biologically appropriate. Instead, endogenous regulation of inserted genes can be achieved through targeting specific sites of integration downstream of the endogenous promoter of the gene of interest.²⁰⁵ The first reported use of homologous DNA donor templates for gene editing in HSCs used zinc finger nuclease to edit the IL-2 receptor common γ chain.²⁰⁶ Follow-up studies²⁰⁷ used CRISPR-Cas9 editing in HSCs.²⁰⁸ A second gene-editing strategy used CRISPR-Cas9 ribonucleoproteins with a chemically modified guide RNA²⁰⁹ and AAV6-mediated homology-directed repair (HDR), with proof of concept for β -hemoglobinopathies.^{210,211} Yet another strategy used CRISPR-Cas9 ribonucleoprotein with linearized single stranded or double stranded DNA templates instead of viral delivery to mediate HDR.⁷⁹ The former was applied to T cells from a patient with a nonsense mutation in the *CD25* gene, where the insertions and deletions alone from nonhomologous end-joining were favorable for CD25 expression and downstream function. These protocols are applicable to other loci as well, although optimal targeting and HDR rates vary for each locus.²¹²

CONCLUSION AND PERSPECTIVES

Studies of patients with IPEX syndrome have paved the way for our understanding of clinical manifestations of Treg cell deficiencies. Function-altering mutations in *FOXP3*, *CTLA4*, *LRBA*, *CD25*, *BACH2*, and *STAT3* revealed the critical role of these proteins in Treg cell function and maintenance of human immune system homeostasis. Studies using patient-derived cells revealed new functions of these genes and elucidated key mechanistic pathways in Treg cell-mediated immune tolerance. Thanks to the rapid technological advances and availability of next-generation sequencing methods, it is reasonable to expect that the number of mutations associated with Treg cell deficiency will continue to grow and that new diseases will soon be added to the existing list of monogenic diseases of immune dysregulation. Characterizing these mutations and the disrupted downstream molecular pathways will be essential to accelerate diagnosis, design new targeted therapies to alleviate disease symptoms, and develop curative approaches such as gene correction.

Because many molecules identified as crucial for Treg cell function and immune tolerance also play a role in other cell types

and physiologic processes, a system-wide comprehension of affected molecular networks in each of these mutations is required to predict short- and long-term side effects of therapeutic interventions in patients. Furthermore, to advance our knowledge and identify cures for monogenic diseases of immune regulation, a multidisciplinary collaborative approach among clinician-scientists in academia and industry is required because patients and samples are rare, clinical phenotypes are often severe, and curative treatments are still at an early stage or only on the horizon.

What is known?

- Clinical phenotypes are largely overlapping for many Tregopathies.
- Whole-exome sequencing or coupling-targeted exon sequencing with mutation-specific functional validation can be used for diagnosing Tregopathies.
- Absolute Treg cell counts determined by using traditional flow cytometry can be inaccurate because FOXP3 is induced during CD4⁺ T-cell activation; TSDR demethylation status should be used for more accurate quantification of FOXP3⁺ Treg cells.

What is still unknown?

- Whether and to what extent pharmacologic treatments, such as rapamycin, can rescue the phenotypes of patients with residual protein function
- Whether HSCT is beneficial in all Tregopathies and whether isolated Treg cell replacement therapy could be beneficial
- Whether Treg cell defects are responsible for autoimmunity detected in other inborn errors of immunity currently not classified as “regulatory T cell defects” by IUIS
- To what extent autoimmune disorders can be rescued by boosting Treg cells numbers, function, or both

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